

SEMINAR 3

Funcții implicite

1. Arătați că ecuația $(x^2+y^2)^3 - 3(x^2+y^2) - 2 = 0$, definește implicit fct. $y = y(x)$ în jurul pct $(1, -1)$.

Calc.: $y'(1), y''(1), T_{\perp}$ asociat fct $y = y(x)$ și pct x .

SOL.

Fie $F: \mathbb{R}^2 \rightarrow \mathbb{R}, F(x, y) = (x^2+y^2)^3 - 3(x^2+y^2) - 2$.

Deriv. parțiale $\begin{cases} \frac{\partial F}{\partial x}(x, y) = 6x(x^2+y^2)^2 - 6x \\ \frac{\partial F}{\partial y}(x, y) = 6y(x^2+y^2)^2 - 6y \end{cases}$

(i) $F \in C^2(\mathbb{R}^2)$ (evident)

(ii) $F(1, -1) \underset{=}{=} 0 : F(1, -1) = (1+1)^3 - 3(1+1) - 2 = 0$

(iii): $\frac{\partial F}{\partial y}(1, -1) \underset{=}{\neq} 0 : -6(1+1)^2 - 6 \cdot (-1) = -24 + 6 = -18 \neq 0$

Atunci, $\exists U \in \mathcal{O}(1), \exists V \in \mathcal{O}(-1)$ și o unică funcție $y = y(x), y: U \rightarrow V$ cu prop.: 1) $y(1) = -1$

2) $F(x, y(x)) = 0, \forall x \in U$ ie $(x^2 + y^2(x))^3 - 3(x^2 + y^2(x)) - 2 = 0$.

3) $y \in C^2(U)$

Formula

$$y'(x) = - \frac{\frac{\partial F}{\partial x}(x, y(x))}{\frac{\partial F}{\partial y}(x, y(x))} =$$

$$= - \frac{6x(x^2 + y^2) - 6x}{6y(x^2 + y^2) - 6y} = - \frac{6x(x^2 + y^2 - 1)}{6y(x^2 + y^2 - 1)}$$

$$= - \frac{x}{y(x)}$$

Calcul direct (calculăm $y'(x)$, derivând ec. diferențială, dar ținând cont că y este funcție de x)

$$(x^2 + y^2)^3 - 3(x^2 + y^2) - 2 = 0$$

$$3(x^2 + y^2)^2 (2x + 2y(x)y'(x)) - 3(2x + 2y(x)y'(x)) = 0 \Rightarrow$$

$$\Rightarrow 6x(x^2 + y^2)^2 + 6y(x)y'(x)(x^2 + y^2)^2 - 6x - 6y(x)y'(x) = 0 \Rightarrow$$

$$\Rightarrow y'(x)(6y(x)(x^2 + y^2)^2 - 6y(x)) =$$

$$= 6x - 6x(x^2 + y^2)^2 \Rightarrow$$

$$\Rightarrow y'(x) = - \frac{x}{y(x)}$$

$$y'(x) = - \frac{1}{y(x)} \frac{dy}{dx} = - \frac{1}{-1} = 1$$

$y''(x) = ?$ ← Calcul direct

$$y''(x) = - \frac{y'(x) - x \cdot y'(x)}{y^2(x)}$$

$$\Rightarrow y''(1) = - \frac{-1 - 2 \cdot 1}{-1 - 2 \cdot 1} = 2$$

$$T_2(x) = y(1) + \frac{1}{1!} y'(1)(x-1) + \frac{1}{2!} y''(1)(x-1)^2$$

$$= -1 + x - 1 + (x-1)^2 = x^2 - x - 1$$

obs

Putem trata ec. si ca o diferentia, astfel:

$$\text{dim ip: } (x^2+y^2)^2 - 2(x^2+y^2) - 2 = 0 \Rightarrow$$

$$\Rightarrow t^2 - 2t - 2 = 0 \Rightarrow (t+1)(t^2 - t - 2) = 0$$

$$\Rightarrow t_1 = -1, t_2 = 2, t_3 = -1$$

$$x^2 + y^2 = -1, \text{ } x, y$$

$$x^2 + y^2 = 2 \Rightarrow y = \pm \sqrt{2 - x^2}$$

$$\text{dim ip: } y(1) = -1$$

$$\Rightarrow y = -\sqrt{2 - x^2}, \text{ } x \in (-\sqrt{2}, \sqrt{2})$$

$$y'(x) = \frac{x}{\sqrt{2 - x^2}} \Rightarrow y'(1) = 1$$

$$y''(x) = \frac{\sqrt{2 - x^2} + x \cdot \frac{-x}{\sqrt{2 - x^2}}}{2 - x^2} = \frac{2}{(2 - x^2)\sqrt{2 - x^2}}$$

$$\Rightarrow y''(1) = \frac{2}{2 \cdot 1} = 1$$

2. Aratati ca ec. $x \cos y + y \cos x + z \cos z - 1 = 0$ defineste implicit

set $z = z(x, y)$ in jurul $(1, 0, 0)$ si

$$\text{calc } \frac{\partial z}{\partial x}(1, 0), \frac{\partial z}{\partial y}(1, 0), \frac{\partial z}{\partial z}(1, 0)$$

sol: Fie $F: \mathbb{R}^3 \rightarrow \mathbb{R}$

$$F(x, y, z) = x \cos y + y \cos x + z \cos z - 1$$

$$\begin{cases} \frac{\partial F}{\partial x} = \cos y - z \sin x \\ \frac{\partial F}{\partial y} = \cos x - x \sin y \\ \frac{\partial F}{\partial z} = \cos x - y \sin z \end{cases}$$

$$(i) F \in \mathcal{C}^1(\mathbb{R}^3)$$

$$(ii) F(1, 0, 0) = \cos 0 - 1 = 0$$

$$(iii) \frac{\partial F}{\partial z}(1, 0, 0) \neq 0 : \frac{\partial F}{\partial z}(1, 0, 0) = \cos 1 \neq 0$$

Atunci $\exists U \in \mathcal{U}((1,0))$, $\exists V \in \mathcal{U}(0)$ și
o unică funcție $z = z(x,y)$, $z: U \rightarrow V$

cu: 1) $z(1,0) = 0$

2) $F(x,y,z(x,y)) = 0$, $\forall (x,y) \in U$
 $\hookrightarrow x \cos y + y \cos z(x,y) + z(x,y) \cos x - 1 = 0$

3) $z \in \mathcal{C}^1(U)$

$\frac{dz}{dx}(x,y) = ?$

$\frac{dz}{dy}(x,y) = ?$

Formule

$\frac{dz}{dx}(x,y) = - \frac{\frac{dF}{dx}(x,y,z(x,y))}{\frac{dF}{dz}(x,y,z(x,y))}$

$\frac{dz}{dy}(x,y) = - \frac{\frac{dF}{dy}(x,y,z(x,y))}{\frac{dF}{dz}(x,y,z(x,y))}$

$\Rightarrow \begin{cases} \frac{dz}{dx}(x,y) = - \frac{\cos y - z(x,y) \sin x}{-y \sin z(x,y) + \cos x} \\ \frac{dz}{dy}(x,y) = - \frac{-x \sin y + \cos z(x,y)}{-y \sin z(x,y) + \cos x} \end{cases}$

Calcul direct în ec. din enunț, derivând
în raport cu x și ținând cont
că z este fct de x și y , $z = z(x,y)$

$x \cos y + y \cos z + z \cos x - 1 = 0 \Rightarrow$
 $\Rightarrow \cos y - y \sin z(x,y) \cdot \frac{dz}{dx}(x,y) +$
 $+ \frac{dz}{dx}(x,y) \cos x + z(x,y) \cdot (-\sin x) = 0 \Rightarrow$

$\Rightarrow \frac{dz}{dx}(x,y) = \dots$

$$\frac{\partial z}{\partial x}(1,0) = - \frac{\cos 0 - z(1,0) \cdot \sin 1}{-0 \cdot \sin 1 + \cos 1} =$$

$$= - \frac{1-0}{\cos 1} = - \frac{1}{\cos 1}$$

$$\frac{\partial z}{\partial y}(1,0) = - \frac{-1 \cdot \sin 0 + \cos z(1,0)}{-0 \cdot \sin 1 + \cos 1} =$$

$$= - \frac{\cos 0}{\cos 1} = - \frac{1}{\cos 1}$$

$$dz(1,0) = \frac{\partial z}{\partial x}(1,0) dx + \frac{\partial z}{\partial y}(1,0) dy =$$

$$= - \frac{1}{\cos 1} (dx + dy)$$

$$\Rightarrow dz(1,0)(u,v) = - \frac{1}{\cos 1} (u+v)$$

3. Det. ecuațiile unei fct $y=y(x)$ definită implicit de ec. $x^3+y^3=2xy$.

Sol:

$$\text{Fie } F: \mathbb{R}^2 \rightarrow \mathbb{R}, F(x,y) = x^3+y^3-2xy.$$

$$\begin{cases} \frac{\partial F}{\partial x} = 3x^2 - 2y \\ \frac{\partial F}{\partial y} = 3y^2 - 2x \end{cases}$$

Ec $F(x,y)=0$ def. implicit fct $y=y(x)$ într-o vecinătate a pct $(y_0, x_0) \in \mathbb{R}^2$ de:

$$1) F \in C^2(\mathbb{R}^2)$$

$$2) F(x_0, y_0) = 0$$

$$3) \frac{\partial F}{\partial y}(x_0, y_0) \neq 0$$

Atunci pte. un punct (x_0, y_0) cu aceste prop., $\exists U \in \mathcal{O}(x_0), \exists V \in \mathcal{O}(y_0)$ si

o univ. fct $y=y(x)$, $y:U \rightarrow V$, at:

1) $y(x_0)=y_0$

2) $F(x, y(x))=0, \forall x \in U$

3) $y \in C^2$

$$y'(x) = - \frac{\frac{\partial F}{\partial x}(x, y(x))}{\frac{\partial F}{\partial y}(x, y(x))} = \frac{2y(x) - 3x^2}{3y^2(x) - 2x}$$

Pro. a determina pct de extrem ale

fct $y=y(x)$, aez sint:

- $\begin{cases} F(x, y) = 0 \\ y'(x) = 0 \\ \frac{\partial F}{\partial y}(x, y) = 0 \end{cases} \Rightarrow$ sol. sint. la putem vedea ca pct (x_0, y_0) cu cond. cantitative
- $y''(x) \rightarrow$ calcul
- $y''(x_0) \begin{cases} + \Rightarrow x_0 - \text{pct de minim local pe fct } y=y(x) \\ - \Rightarrow x_0 - \text{pct de maxim local} \end{cases}$

la ma:

$$\begin{cases} F(x, y) = 0 \\ y'(x) = 0 \\ \frac{\partial F}{\partial y}(x, y) \neq 0 \end{cases} \Rightarrow \begin{cases} x^3 + y^3 - 2xy = 0 \\ -3x^2 - 2y = 0 \\ 3y^2 - 2x = 0 \end{cases}$$

$$y = \frac{3x^2}{2} \Rightarrow x^3 + \left(\frac{3x^2}{2}\right)^3 - 2x \cdot \frac{3x^2}{2} = 0 \Rightarrow$$

$$\Rightarrow x^3 - \frac{27x^6}{8} - 3x^3 = 0 \Rightarrow$$

$$\Rightarrow -2x^3 - \frac{27x^6}{8} = 0 \Rightarrow$$

$$\Rightarrow x^3 \left(-2 - \frac{27x^3}{8}\right) = 0$$

$$\Rightarrow x = 0 \text{ sau } x^3 = -\frac{16}{27}$$

$$\Rightarrow y = 0$$

$$\Rightarrow (0, 0) \text{ nu verifica (*)}$$

$$\Rightarrow x = -\frac{2\sqrt[3]{16}}{3}$$

$$\Rightarrow y = \frac{2\sqrt[3]{16}}{3} \Rightarrow \left(-\frac{2\sqrt[3]{16}}{3}, \frac{2\sqrt[3]{16}}{3}\right) \text{ verifica (*)}$$

Pentru $\left(-\frac{2\sqrt{2}}{3}, \frac{2\sqrt{14}}{3}\right)$ poate fi văzut că
 (20, y0) nu cond. anterioră $\Rightarrow$
 avem def. implicită $y = y(x)$ cu pt.
 critic $-\frac{2\sqrt{2}}{3}$ și $y\left(-\frac{2\sqrt{2}}{3}\right) = \frac{2\sqrt{14}}{3}$.

Cum $y'(x) = -\frac{3x^2 - 2y(x)}{3y^2(x) - 2x} \Rightarrow$

$$\Rightarrow y''(x) = \left(\frac{-3x^2 + 2y(x)}{3y^2(x) - 2x} \right)' = \dots$$

BAU

Cum $y'(x) = -\frac{3x^2 + 2y(x)}{3y^2(x) - 2x} \Rightarrow$

$$\Rightarrow y'(x) \cdot 3y^2(x) - 2xy'(x) = -3x^2 + 2y(x)$$

$$\Rightarrow y''(x) \cdot 3y^2(x) + y'(x) \cdot 6y(x) \cdot y'(x) -$$

$$-2y'(x) - 2xy''(x) = -6x + 2y'(x) \Rightarrow$$

$$\Rightarrow y''(x) = \frac{-6x + 4y'(x) - y'(x)^2 6y(x)}{3y^2(x) - 2x}$$

$$\Rightarrow y''\left(-\frac{2\sqrt{2}}{3}\right) = \frac{-6 \cdot \left(-\frac{2\sqrt{2}}{3}\right)}{3 \cdot \left(\frac{2\sqrt{14}}{3}\right)^2 - 2 \cdot \left(-\frac{2\sqrt{2}}{3}\right)}$$

$y'\left(-\frac{2\sqrt{2}}{3}\right) = 0$
 $y\left(-\frac{2\sqrt{2}}{3}\right) = y_0 = \frac{2\sqrt{14}}{3}$

$\Rightarrow -\frac{2\sqrt{2}}{3}$ este punct de minim local
 pt. funcția $y = y(x)$ def. implicit
 de ecuația dată.

4. Examinăm funcția de 2 pt $y = y(x)$ def. implicit de ec. $x^3 + y^3 = 3xy + 3$.

Sol:

$$\text{Fie } F: \mathbb{R}^2 \rightarrow \mathbb{R}, F(x, y) = x^3 + y^3 - 3xy + 3$$

$$\frac{\partial F}{\partial x}(x, y) = 3x^2 - 3y$$

$$\frac{\partial F}{\partial y}(x, y) = 3y^2 - 3x$$

Ec. $F(x, y) = 0$ def. implicit 2 pt $y = y(x)$ într-o vecinătate a pt $(x_0, y_0) \in \mathbb{R}^2$ dacă:

$$1) F(x_0, y_0) = 0$$

$$2) \frac{\partial F}{\partial y}(x_0, y_0) \neq 0$$

$$3) F \in C^2.$$

Dacă (x_0, y_0) este un punct cu aceste prop., atunci $\exists U \in \mathcal{O}(x_0)$, $V \in \mathcal{O}(y_0)$ și o unică funcție $y = y(x)$, $y: U \rightarrow V$ at:

$$1) y(x_0) = y_0$$

$$2) F(x, y(x)) = 0, \forall x \in U$$

$$3) y \in C^2$$

$$y'(x) = ?$$

formulă

$$y'(x) = - \frac{\frac{\partial F}{\partial x}(x, y(x))}{\frac{\partial F}{\partial y}(x, y(x))} = - \frac{3x^2 - 3y(x)}{3y^2(x) - 3x} = \frac{x^2 - 2xy(x)}{x^2 - y^2(x)}, \forall x \in U$$

calcul direct apăsând

$$\begin{cases} F(x,y) = 0 \\ y'(x) = 0 \\ \frac{\partial F}{\partial y}(x,y) \neq 0 (*) \end{cases} \Rightarrow \begin{cases} x^3 + y^3 - 3x^2y - 3 = 0 \\ x^2 - 2xy = 0 \\ 3y^2 - 3x^2 \neq 0 \end{cases} \Rightarrow$$

$$\Rightarrow x(x-2y) = 0 \begin{cases} x = 0 \text{ (I)} \\ x = 2y \text{ (II)} \end{cases}$$

(I) $x=0 \Rightarrow y^3=3 \Rightarrow y=\sqrt[3]{3} \Rightarrow (0, \sqrt[3]{3})$ (*) verif.

(II) $x=2y \Rightarrow 9y^3 - 12y^3 - 3 = 0 \Rightarrow y^3 = 1 \Rightarrow y=1$
 $\Rightarrow x=2$ (*) verif
 $(-2, -1)$

Sol. sist. sunt $(0, \sqrt[3]{3})$ și $(-2, -1)$ ce pot fi văzute ca pct. de extrema (x_0, y_0) cu prop. anterioare.

→ caup. pct $(0, \sqrt[3]{3})$ avem pct. am-
pliată $y_1 = y_1(x)$ cu pct. critic 0
și $y_1(0) = \sqrt[3]{3}$

→ caup. pct $(-2, -1)$ avem pct. am-
pliată $y_2 = y_2(x)$ cu pct. critic -2
și $y_2(-2) = -1$.

$$y_i''(x) = ?, i = 1, 2$$

$$\text{Știm } y_i'(x) = \frac{x^2 - 2x \cdot y_i(x)}{x^2 - y_i^2(x)} \Rightarrow$$

$$\Rightarrow y_i'(x) \cdot x^2 - y_i'(x) \cdot y_i^2(x) = x^2 - 2x y_i(x) \Rightarrow$$

$$\Rightarrow y_i''(x) \cdot x^2 + 2x y_i'(x) - y_i''(x) \cdot y_i^2(x) -$$

$$- 2y_i(x) \cdot (y_i'(x))^2 = 2x - 2y_i(x) - 2x y_i(x)$$

$$\Rightarrow y_i''(x) (x^2 - y_i^2(x)) = -4x y_i'(x) + 2y_i(x) y_i'(x)^2 + 2x - 2y_i(x) \Rightarrow$$

$$\Rightarrow y_1''(x) = \frac{-4xy_1'(x) + 2y_1(x)(y_1'(x))^2}{x^2 - y_1^2(x)}$$

Deci:

pta fct $y_1 = y_1(x)$:

$$y_1''(0) = \frac{0+0-2\sqrt{3}}{0-(\sqrt{3})^2} = \frac{2}{3} > 0$$

$$y_1'(0) = 0$$

$$y_1(0) = \sqrt{3}$$

$\Rightarrow$ pnt $x=0$ este punct de minim

local pta fct $y_1 = y_1(x)$ def implicit
în jurul lui $(0, \sqrt{3})$

pta fct $y_2 = y_2(x)$:

$$y_2''(-2) = \frac{2 \cdot (-2) - 2 \cdot (-1)}{(-2)^2 - (-1)^2} = -\frac{2}{3} < 0$$

$$y_2'(-2) = 0$$

$$y_2(-2) = -1$$

$\Rightarrow$ pnt $x=-2$ este punct de maxim

local pta fct $y_2 = y_2(x)$ def implicit
în jurul lui $(-2, -1)$

5. Det. extremele fct implicite $F = F(x, y, z)$
def. de ec. $5x^2 + 5y^2 + 5z^2 - 2xy - 2xz - 2yz - z^2 = 0$.

Sol:

$$F \text{ pe } \overline{F}: \mathbb{R}^3 \rightarrow \mathbb{R}, F(x, y, z) = 5x^2 + 5y^2 + 5z^2 - 2xy - 2xz - 2yz - z^2 = 0$$

$$\begin{cases} \frac{\partial F}{\partial x}(x, y, z) = 10x - 2y - 2z \\ \frac{\partial F}{\partial y}(x, y, z) = 10y - 2x - 2z \\ \frac{\partial F}{\partial z}(x, y, z) = 10z - 2x - 2y \end{cases}$$

Ec. $F(x, y, z) = 0$ def. implicit F
 $z = z(x, y)$ într-o vecinătate a unui
 pt. $(x_0, y_0, z_0) \in \mathbb{R}^3$ dat:

$$\begin{cases} F(x_0, y_0, z_0) = 0 \\ \frac{dF}{dz}(x_0, y_0, z_0) \neq 0 \\ F \in C^2 \end{cases}$$

Atunci pt. un punct (x_0, y_0, z_0)
 cu aceste prop., $\exists U \in \mathcal{O}((x_0, y_0))$,
 $\exists V \in \mathcal{O}(z_0)$ și o unică F , $z = z(x, y)$,
 $z: U \rightarrow V$ at

$$\begin{cases} z \in C^2 \\ F(x_0, y_0, z_0) = z_0 \\ F(x, y, z(x, y)) = 0, \forall (x, y) \in U \\ \frac{dF}{dx}(x, y, z(x, y)) \end{cases}$$

$$\begin{cases} \frac{dz}{dx}(x, y) = - \frac{\frac{dF}{dx}(x, y, z(x, y))}{\frac{dF}{dz}(x, y, z(x, y))} \\ \frac{dz}{dy}(x, y) = - \frac{\frac{dF}{dy}(x, y, z(x, y))}{\frac{dF}{dz}(x, y, z(x, y))} \end{cases}$$

$$\Rightarrow \begin{cases} \frac{dz}{dx}(x, y) = - \frac{10x - 2y - 2z(x, y)}{10z(x, y) - 2x - 2y} \\ \frac{dz}{dy}(x, y) = - \frac{10y - 2x - 2z(x, y)}{10z(x, y) - 2x - 2y} \end{cases}$$

Pt. a det. punctele de extrem. $z(x, y)$
 rest:

$$\begin{cases} F(x, y, z) = 0 \\ \frac{dz}{dx}(x, y) = 0 \\ \frac{dz}{dy}(x, y) = 0 \\ \frac{d^2F}{dz^2}(x, y, z) \neq 0 \end{cases} \Rightarrow \begin{cases} 3x^2 + 5y^2 + 5z^2 - 2xy - 2xz - 2yz = 0 \\ 10x - 2y - 2z = 0 \\ 10y - 2x - 2z = 0 \\ 10z - 2x - 2y \neq 0 \end{cases} \textcircled{*}$$

$$\Rightarrow 12x - 12y = 0 \Rightarrow x = y$$

$$10x - 2x - 2z = 0 \Rightarrow z = 4x$$

$$5x^2 + 5x^2 + 5 \frac{(4x)^2}{80x^2} - 2x^2 - 2x \cdot 4x - 2x \cdot 4x - 72 = 0 \Rightarrow$$

$$\Rightarrow 80x^2 - 18x^2 - 72 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\textcircled{I} \quad x = 1 \Rightarrow y = 1, z = 4 \Rightarrow (1, 1, 4) \text{ verif. } (*)$$

$$\textcircled{II} \quad x = -1 \Rightarrow y = -1, z = -4 \Rightarrow (-1, -1, -4) \text{ verif. } (*)$$

Sol. intermediei sunt $(1, 1, 4)$ și $(-1, -1, -4)$ ce pot fi văzute ca pct. de forma (x_0, y_0, z_0) cu cond. anterioare.

Deci pct. pct. $(1, 1, 4)$, $\exists$ pct. implicată $\Xi_1 = \Xi_1(x, y)$ cu pct. critic $(1, 1)$ și $\Xi_1(1, 1) =$

$\rightarrow$ pct. pct. $(-1, -1, -4)$, $\exists$ pct. implicată $\Xi_2 = \Xi_2(x, y)$ cu pct. critic $(-1, -1)$ și $\Xi_2(-1, -1) = -4$

$$\text{Cum } \frac{\partial \Xi_i}{\partial x}(x, y) = - \frac{10x - 2y - 2\Xi_i(x, y)}{10\Xi_i(x, y) - 2x - 2y} =$$

$$= \frac{y + \Xi_i(x, y) - 5x}{5\Xi_i(x, y) - x - y}$$

$$\frac{\partial \Xi_i}{\partial y}(x, y) = \frac{\Xi_i(x, y) + x - 5y}{5\Xi_i(x, y) - x - y}$$

$$\frac{\partial^2 \Xi_i}{\partial x^2}(x, y) = \frac{1}{\partial x} \left(\frac{\partial \Xi_i}{\partial x}(x, y) \right) =$$

$$= \frac{\left(\frac{\partial \Xi_i}{\partial x}(x, y) - 5 \right) (5\Xi_i(x, y) - x - y) - (y + \Xi_i(x, y) - 5x) \frac{\partial \Xi_i}{\partial x}(x, y)}{(5\Xi_i(x, y) - x - y)^2}$$

$$= \frac{\left(1 + \frac{\partial \Xi_1}{\partial y}(x, y)\right) \left(5\Xi_1(x, y) - x - y\right) - (-5x + y + \Xi_1(x, y)) \left(\frac{\partial \Xi_1}{\partial y}\right)}{(5\Xi_1(x, y) - x - y)^2}$$

$$\frac{\partial^2 \Xi_1(x, y)}{\partial y^2} = \frac{(-5 + \frac{\partial \Xi_1}{\partial y})(5\Xi_1 - x - y) - (\Xi_1 + x - 5y) \left(\frac{\partial \Xi_1}{\partial y}\right)}{(5\Xi_1(x, y) - x - y)^2}$$

Deci, pt $\Xi_1 = \Xi_1(x, y)$:

$$\frac{\partial \Xi_1}{\partial x}(1, 1) = 1 + 4 - 5 = 0$$

$$\frac{\partial \Xi_1}{\partial y}(1, 1) = \frac{4 + 1 - 5}{5 \cdot 4 - 1 - 1} = 0$$

$$\frac{\partial^2 \Xi_1}{\partial x^2}(1, 1) = \frac{-5(5 \cdot 4 - 1 - 1) - (-5 + 1 + 4)(-1)}{(5 \cdot 4 - 1 - 1)^2} =$$

$$= -\frac{5}{18}$$

$$\frac{\partial^2 \Xi_1}{\partial x \partial y}(1, 1) = \frac{\partial^2 \Xi_1}{\partial y \partial x}(1, 1) = \frac{5 \cdot 4 - 1 - 1 - (-5 + 1 + 4)(-1)}{18^2} = \frac{1}{18}$$

$$\frac{\partial^2 \Xi_1}{\partial y^2}(1, 1) = \frac{-5(5 \cdot 4 - 1 - 1) - (4 + 1 - 5)(-1)}{18^2} = -\frac{5}{18}$$

$$H_{\Xi_1}(1, 1) = \begin{pmatrix} -\frac{5}{18} & \frac{1}{18} \\ \frac{1}{18} & -\frac{5}{18} \end{pmatrix}$$

$$\Delta_1 = -\frac{5}{18} < 0, \Delta_2 = \frac{25}{18^2} - \frac{1}{18^2} > 0 \Rightarrow$$

$\Rightarrow (1, 1)$ - punct de maxim local

pt $\Xi_2 = \Xi_2(x, y)$:

$$\frac{\partial \Xi_1}{\partial x}(-1, -1) = -1 - 4 + 5 = 0$$

$$\frac{\partial \Xi_1}{\partial y}(-1, -1) = 0$$

$$H_{\Xi_2}(-1, -1) = \begin{pmatrix} \frac{5}{18} & -\frac{1}{18} \\ -\frac{1}{18} & \frac{5}{18} \end{pmatrix}$$